Rudraksh Tiwari

+91 9741973906 | rudraksh78779@gmail.com | Bengaluru, Karnataka
linkedin.com/in/rudraksh-tiwari-rt | github.com/Rudraksh-Tiwari | rudraksht.tech

SUMMARY

Final-year Computer Science and Engineering undergraduate (graduating June 2027) focused on backend engineering, reliability, and correctness under concurrency. Built a distributed task queue from scratch on Redis with zero job loss proven under chaos testing and 57 tests in CI, and shipped a live fintech product (FastAPI, PostgreSQL, AWS) that processes real bank statement data. Comfortable learning unfamiliar codebases and languages quickly. Seeking a Software Engineer Intern role building production systems.

EDUCATION

Nitte Meenakshi Institute of Technology, Bengaluru 2023 – 2027
Bachelor of Engineering, Computer Science and Engineering CGPA: 8.32 / 10

- Coursework: Data Structures, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Deva Matha Central School, Bengaluru 2021 – 2023
Class 12 (CBSE): 76.6% | Class 10: 81.1%

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL
Backend: REST API Design, FastAPI, Spring Boot, JWT Authentication, Background Job Processing, Concurrency, Distributed Systems
Databases: PostgreSQL (schema design, migrations), MySQL, MongoDB, Redis, ChromaDB
Infrastructure & Tools: Linux, Docker, AWS EC2, nginx, TLS, Git, GitHub Actions (CI), Prometheus, React
Core CS: Data Structures, Algorithms, Complexity Analysis, Object-Oriented Design, Operating Systems, DBMS, Computer Networks

PROJECTS

Reliable Distributed Task Queue | Python, Redis, Lua, FastAPI, Docker, Prometheus 2025 – 2026

- Built a distributed task queue from scratch in Python and Redis (no Celery/RQ): atomic job claims via Lua scripts, lease-based recovery of crashed workers' jobs, capped exponential backoff with jitter, priority lanes, delayed jobs, idempotency keys, and a dead-letter queue.
- Proved zero job loss under failure with a chaos test that hard-kills workers mid-job (600/600 jobs completed, none lost); sustains ~2,000 jobs/sec on a laptop and 3,700+/sec in a Linux container with 4 workers × 8 threads.
- Wrote 57 unit, integration, and concurrency tests against real Redis (20 threads racing for one job, concurrent reapers, simultaneous replays) and a GitHub Actions CI pipeline running crash-recovery and chaos proofs on every push; found and fixed a replay/discard race that silently lost jobs, writing the failing test first.
- Built a FastAPI ops dashboard and hand-written Prometheus /metrics endpoint backed by fleet-wide Redis counters and a p95 latency histogram, with dead-letter replay/discard; closed an XSS hole where job data was rendered as HTML.

Clarifi: AI Financial Copilot | FastAPI, React, PostgreSQL, ChromaDB, AWS | clarifi.page Live, 2026

- Built and deployed a full-stack fintech application on AWS EC2 behind nginx with TLS, designing REST APIs with JWT authentication and per-user data isolation, and owning PostgreSQL schema design and migrations.
- Built an ingestion pipeline for real bank statement exports, handling dynamic header detection, separate debit/credit columns, and day-first date parsing across inconsistent source formats.
- Root-caused and rebuilt a flawed transaction anomaly detector using log-scaled per-category robust z-scores (median/MAD), cutting false positives from 13 to 5 explained flags on the same statement.

Credit Risk Decisioning Engine | Python, XGBoost, SHAP, Streamlit | GitHub 2024

- Built an end-to-end credit risk scoring system on 45,000+ loan records, reaching ~0.978 AUC-ROC with class-imbalance correction, and used SHAP to explain individual approval and rejection decisions.

Early Diabetes Detection (Major Project) | Python, ML | Team of students 2026 – 2027

- Collaborating in a multi-person team under a faculty guide; presenting design and progress at formal project reviews and incorporating feedback from the guide and review panel.

EXPERIENCE

Web Development Intern, SkillCraft Technology, Bengaluru Oct 2024 – Nov 2024

- Built frontend and backend features with HTML, CSS, and JavaScript within a real project structure, using Git version control and team development workflows.

ACHIEVEMENTS AND CERTIFICATIONS

- Runner-Up, Coding Battles inter-college competitive programming contest (2024)
- Data Structures and Algorithms using Java, Web Development: Apna College (2025)
- MongoDB Python Developer Path and Database Administration Path (2025); Infosys Springboard: Java Foundation, Software Engineering Fundamentals, Machine Learning Foundation